

Diversity Healthcare

Phase 1 Project Proposal

Contributed by:

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Team Expectation Agreement 1

Project Leader: Anabelle Hsiao

Assigned Communicator: Muhammad Bilal

1. Methods of Communication

- Primary method: Group chat (WhatsApp/Discord/Slack) for quick updates.
- Secondary method: Email for formal communications and sharing files.
- Emergency/urgent matters: Phone/text messages.

2. Communication Response Times

- Group chat messages: within 12 hours.
- Emails: within 24 hours.
- Urgent matters (labeled as urgent): within 2 hours, when possible.

3. Regular Meetings

- Frequency: Weekly scheduled meetings (day/time agreed by team).
- Additional meetings may be called if needed.
- All meetings are mandatory, unless excused in advance.

4. Running Meetings

- Meetings will be held online (Zoom/Teams) unless otherwise agreed.
- A rotating facilitator will run the meeting.
- A designated note-taker (rotating each week) will keep minutes and share them in the group chat.

5. Meeting Preparations

- Members are expected to review the agenda and prepare relevant updates.
- Assigned tasks must be completed before the meeting to allow progress review.

6. Division of Work

- Work will be divided equally based on strengths, interests, and availability.
- Decisions will be made through team consensus.
- If disagreement occurs, the majority vote will decide.

7. Submitting Work

- Each member must submit their portion of work at least 24 hours before deadlines for review.
- A designated editor/compilation lead will review, integrate, and submit final work.
- All team members must approve the final submission before it is sent.

8. Contingency Planning

- If a team member drops out, work will be redistributed equally among remaining members.
- If a member is sick or unavailable for an extended time, they must inform the team immediately.
- If a member consistently misses meetings or fails to complete work, the team will:
 1. Address the issue directly and respectfully.
 2. Reassign work if necessary.
 3. Contact the TA if the problem continues.
- In the case of academic dishonesty, the team will immediately notify the TA.

9. Additional Guidelines

- Be respectful of each other's ideas and contributions.
- Resolve conflicts professionally and constructively.
- Everyone is encouraged to ask for help if they are struggling with their assigned work.

Proposal

The problem our team chose to solve is the lack of unified systems in digital health. Clinicians (Medical information/record/imaging) are faced with fragmented medical platforms across many systems, leading to delays in diagnosis and errors in treatment. Patients also lack easy access to their health information, making it difficult to keep track of treatments or communicate effectively with clinicians. A secure, user-friendly medical application that unifies records, streamlines collaboration, and empowers patients to be active participants in their care is required.

The invisibility of this problem is important because fragmentation leads to well-intentioned actions that sometimes have the unintended consequence of making things worse. These unintended consequences include the following. Inefficiency: delayed medical care. Ineffectiveness: time-consuming in collecting data across boards, leading to inefficient progress for treating patients or conducting medical research. Inequality: with the lack of a unified system, people in rural areas are often more likely to get misdiagnosed due to the lack of resources and needing to fetch multiple platforms' information to secure appointments, diagnoses, etc. On the other hand, lower-income households will face the same issue due to the lack of financial stability.

Target Audience includes patients (including elders, parents who are booking for minors, patients with disabilities, adults) and medical workers (including doctors, nurses, clinic/hospital workers)

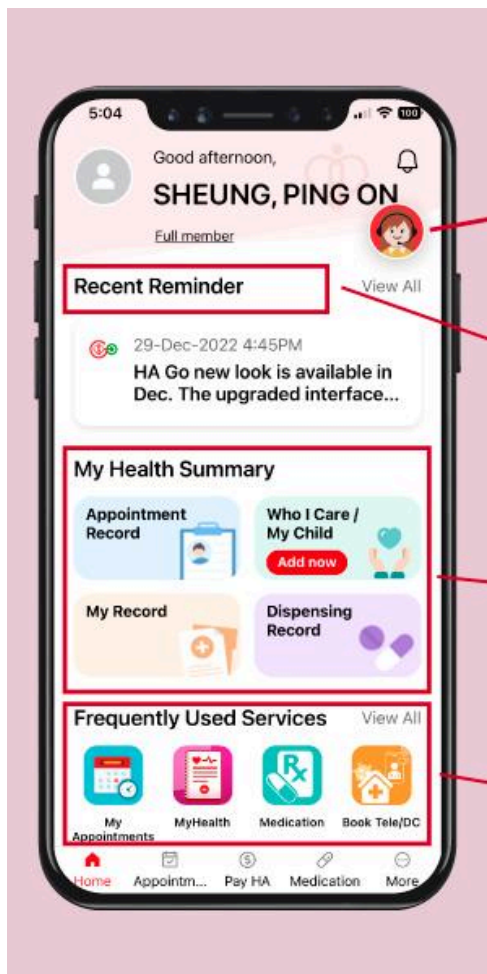
Our Application requirements for our solution include a few factors. First, mobile applications for patients end with simplified information (diagnosis/prescription/appointment(s)). Second, a web version for desktop (so clinicians/patients can enter/view/adjust data). Third, user-friendly applications, specifically for patients who have disability issues such as high colour contrast, size-adjustable buttons, and descriptive text alternatives for images. Fourth, region-based information integration, i.e. the user can add their provincial health card and OHIP / insurance status for Americans, so clinicians have information ahead of time. This allows state/province platform integration (to connect with existing medical records).

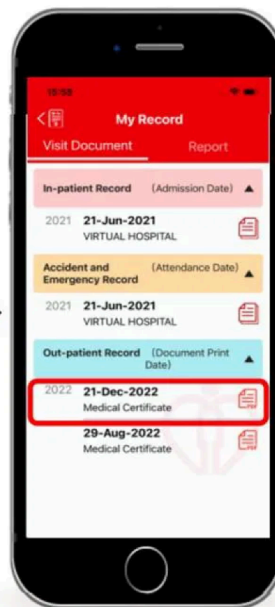
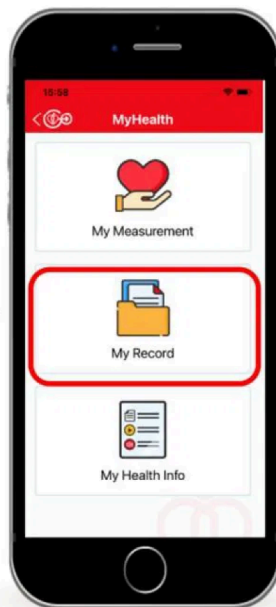
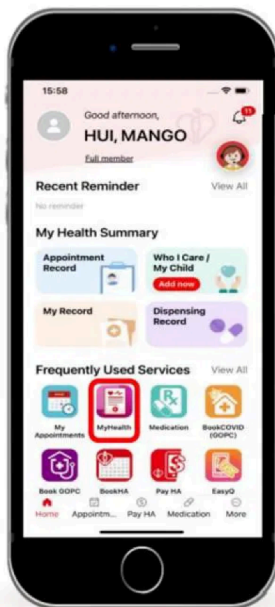
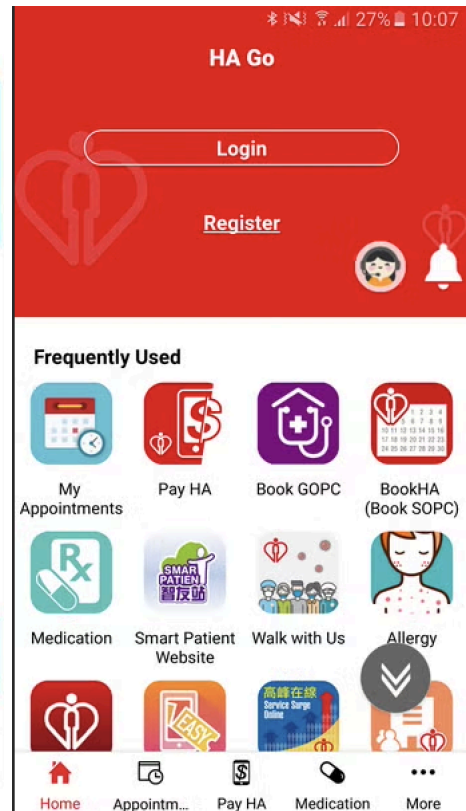
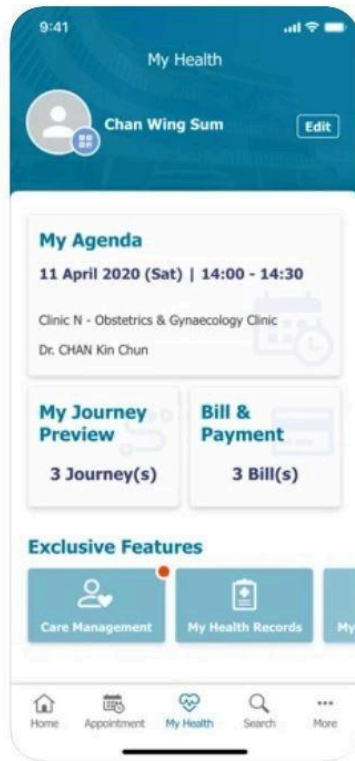
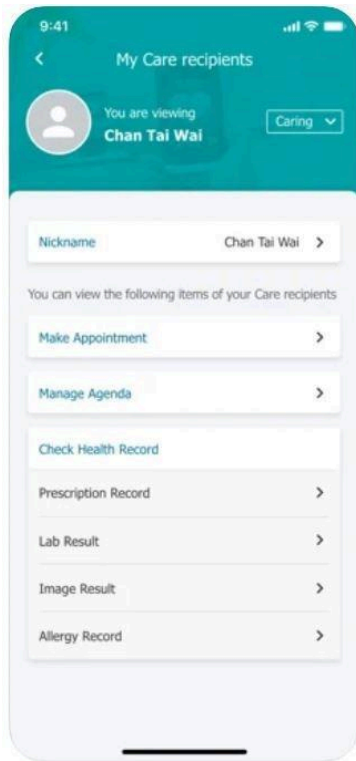
Appendix

Examples and ScreenShots

eHealth/HA Go (Government Healthcare App in Hong Kong)

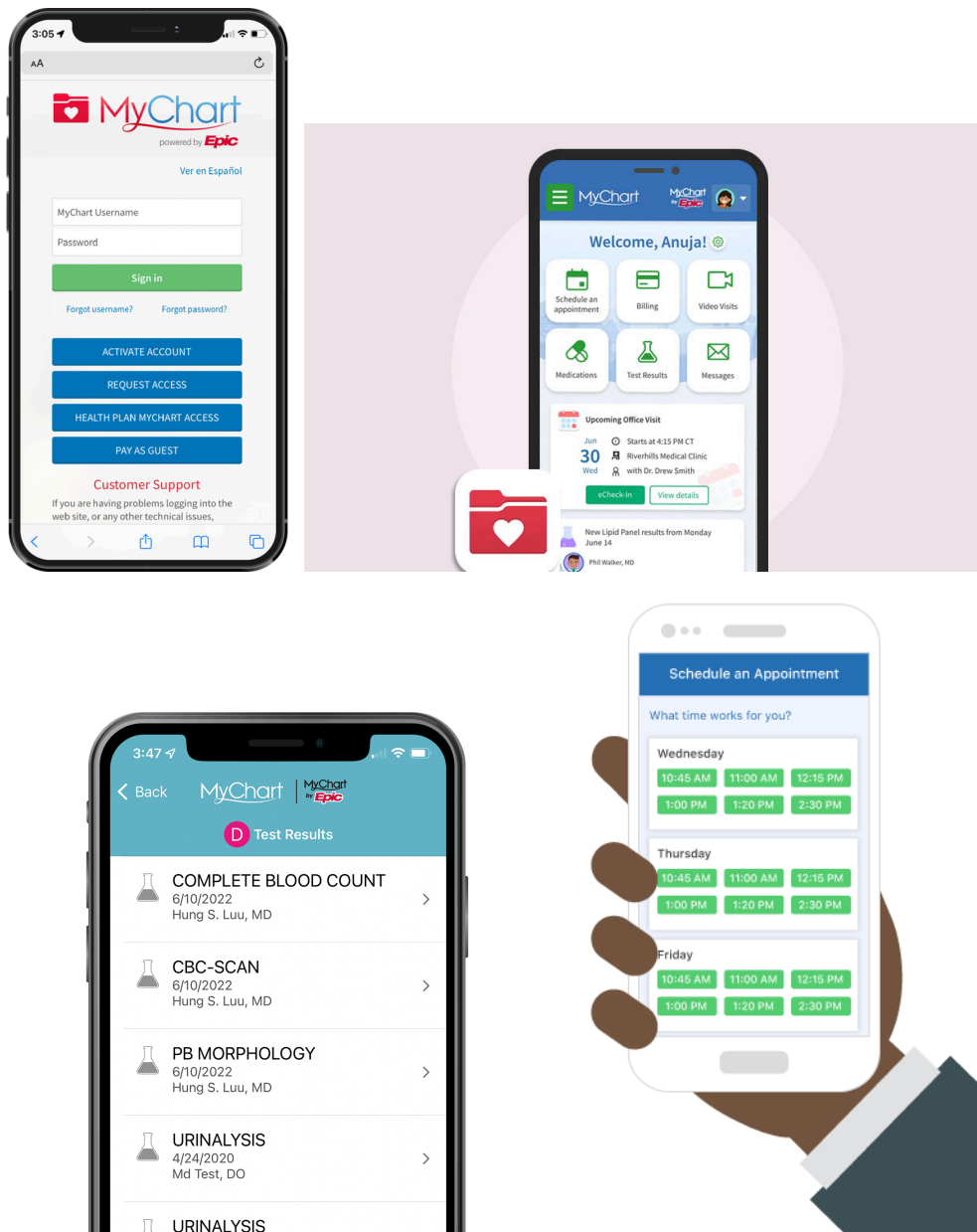
- Mobile app that records patient's healthcare information like diagnosis, prescription, appointment(s), medical bills, etc
- User-friendly, high colour contrast
- Region-based information (registers your region and id)





MyChart :

- Mobile and web app that provides patients access to their healthcare information, including test results, prescriptions, appointment scheduling, visit summaries, and secure messaging with providers.
- User-friendly, but the design varies by hospital/clinic implementation.
- Hospital-based information (linked to a specific health network; patients need separate logins if they use multiple systems)



Literature Review(support)

a) *Accessibility and Usability of State Health Department COVID-19 Vaccine Websites (Howe et al., 2021)*

The article rolls out platform usability issues for US covid 19 vaccine websites (Howe et al., 2021). Though all websites were viewable by computer, 14% did not enable complete smartphone viewability, and 18% were English only. The mean (SD) web accessibility score was 82% . 94% presented eligibility information with median (range) readability scores of 15.6 (9.5-38.9), and most did so at a post-high-school reading level. Usability evaluation showed that 30 states had no indicator of when information was last updated. Of the 47 states providing eligibility information, 57% did not support verifying eligibility. 25 states (50%) provided web-based appointment scheduling, but the majority, 23 out of 25 [92%] required the user to search multiple vaccine locations individually for availability, and only 3 states (6%) had a wait-list or follow-up option when no appointments were available. Nine states (18%) did not offer web-based scheduling or an option for a wait-list or follow-up.

Based on the above statistics, Usability for health platforms are still underdeveloped and are required for integration in subsections such as readability, viewability on smartphones, follow-up notifications, and more. This article further proves the problem of fragmented platforms for health systems and the lack of accessibility on clinical websites. Though the article specifically examined Covid-19 vaccine websites, its findings are broadly applicable to other areas of digital health. The lack of usability highlighted by Howe et al. (2021) reflects a systemic issue across the healthcare community, making it essential to address such barriers to access.

b) *Health Care Professionals' Perspectives on Implementing Patient-Accessible Electronic Health Records in Primary Care: Qualitative Study (Muli et al., 2025)*

This qualitative study examines clinicians' experiences after introducing patient-accessible electronic health records (PAEHRs), which allow patients online record access (ORA) to their medical notes and results in Swedish primary care. Fourteen health care professionals (HCPs) from diverse roles were interviewed six months after launch. Using the Consolidated Framework for Implementation Research (CFIR), the authors identified barriers across technology, organization, people, and process. HCPs described the underlying electronic health record (EHR) as complex, with inconsistent templates, limited safeguards for sensitive data, and uncertainty about patient access. They also reported high workloads, uneven documentation practices, and fragmented communication during implementation. Concerns included exclusion of patients with low digital literacy, older adults, and migrants, as well as increased clinician workload and documentation changes that reduced clinical usefulness.

Facilitators included existing documentation structures, some platform flexibility (e.g. hiding in-progress notes), and patient-engagement practices. The authors conclude that usability, clear organizational processes, and education for both HCPs and patients are essential for equitable access. The findings provide concrete evidence of

fragmentation, inefficiency, and inequity in current digital health systems, aligning directly with our problem statement on the lack of unified, inclusive platforms.

c) *The influence of electronic health record use on collaboration among medical specialties*
(Vos et al., 2020)

The article analyzes the dual role of electronic health records (EHRs) in promoting and hindering collaboration among healthcare practitioners. By integrating patient data and facilitating shared decision-making, the study identifies six collaborative affordances of EHRs that support clinicians in coordinating care: portability, co-located access, shared overview, mutual awareness, messaging, and orchestration. Strict authorization procedures, information overload, and specialty-specific interfaces, however, frequently limit cooperation and lower process effectiveness. The study further demonstrates that while fewer in-person interactions are meant to save time, they can actually restrict the development of trust and efficient workflow, and that challenges in exchanging medical records with outside providers worsen continuity of care.

These findings strongly endorse the necessity for a cohesive, user-focused system as outlined in our proposal, one that not only integrates records across platforms but also mitigates issues related to accessibility, usability, and communication. In addition, the paper emphasizes that in order to fully realize the collaborative potential of EHR systems, organization-wide policies and coordinated use are necessary, highlighting the significance of thoughtful design decisions and unifying these platforms. Hence, without deliberate design and coordinated implementation, fragmentation continues, compromising efficiency, effectiveness, and fairness in healthcare delivery.

Literature Review(solution)

- a) *Mobile Healthcare and People with Disabilities: Current State and Future Needs (Jones et al., 2018)*

The article argues that developers should build every healthcare mobile app for accessibility from the start. Mobile health services grow fast, yet disabled people still appear only in small numbers among both creators and users. More than eighty percent of U.S residents who have disabilities also live with a chronic illness and receive less preventive care, record high obesity rates, exercise less and smoke more often. Data show that seventeen percent of disabled adults use a health related mobile app, while thirty four percent of adults without a disability do so. Tests of current apps show that most excluded people who are blind, have limited hand movement or live with cognitive impairment — none of the studied programs gave blind participants full access and many disabled users report anger because the software lacks the functions they need. This indicates a gap in the industry for mobile health apps that focus on accessibility for the subset of the population most inclined to need it.

- b) *Impact of Patient Access to Their Electronic Health Record: Systematic Review (Tapuria et al., 2021)*

The article reviews over seventy studies on how giving patients access to their own electronic health records (EHRs) affects care. Most research found benefits such as reduced anxiety, stronger doctor–patient relationships, better medication adherence, improved outcomes like blood pressure and blood sugar control, and greater engagement in self-management. Patients also reported being better prepared for consultations and more confident in their care decisions. However, the review noted concerns including privacy, security, and the risk of anxiety when patients view sensitive results online. Older adults and people with low digital literacy were less likely to use portals, highlighting adoption challenges. Overall, the article concludes that patient access to EHRs is a key step toward more sustainable, patient-centered healthcare. This supports our proposal for a unified and accessible app, showing that when patients can view and manage their records, outcomes improve and trust grows, but the design must also prioritize security, equity, and usability

c) *Patient-Centered Digital Health Records and Their Effects on Health Outcomes: Systematic Review (Martijn R Brands et al., 2022)*

This systematic review analyzed studies evaluating patient-centered digital health records and their impact on care. The findings showed that when patients have clear access to their health information through user-friendly platforms, outcomes improved in several ways: patients received more recommended care services, gained better disease knowledge, adhered more consistently to treatments, and demonstrated higher self-management and confidence in their care. Some clinical outcomes, especially for chronic conditions like diabetes and cardiopulmonary disease, also improved. The study emphasizes that patient-centered digital records enhance empowerment and engagement, supporting the idea that unifying fragmented health data into an accessible platform can reduce errors, delays, and frustrations for patients while improving the quality of care.

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